

MTH202-Discrete Mathematics

Solved MCQS for Midterms papers

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LMS Handling

Subject Enrollment

- 1) Online classes available
HTML, CSS, Javascript
Python..etc
- 2) Solved Graded Activities
 - Assignment's
 - Quizzes
 - GDB's
- 3) Solved Quiz File's
- 4) Short Note's
- 5) Past paper's &
Current paper's

Website Link

vulmshelp.com

Final project CS619

- 1) SRS
(Software Requirement's
Specification)
- 2) DD
(Design Document)
- 3) Test phase + viva
- 4) Viva preparation
- 5) Final Deliverable

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Range of function $f(x) = (e^x)$ is _____

- Set of positive real numbers

Composite relation symbolically written as _____

- $S \circ R = \{(a,c) \mid a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$

If $x \equiv 17 \pmod{5}$ which of the following integers are valid solution for x ?

- 12

Range of the relation $\{(0,1), (3,22), (90,34)\}$

- $\{1,22,34\}$

Let $A = \{0,1,2\}$ and $R = \{(0,2), (1,1), (2,0)\}$ be a relation on A . The which of the following ordered pairs are needed to make it transitive?

- $(0,0)$ and $(2,2)$

Operation of subtraction is a binary operation on the set of _____

- Integers

Let $S = R$ and define the 'square' relation $R = \{(x,y) \mid x^2 = y^2\}$. The square relation is an _____ relation

- Equivalence relation

The logic gate NOT is a unary operation on $\{0,1\}$.

- True

Let $A = \{1,2\}$, then $P(A) =$ _____

- $\{\emptyset, \{1\}, \{2\}, \{1,2\}\}$

If a relation R is reflexive, anti symmetric and transitive then which of the following is not true for the inverse relation.

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- Inverse relation will be irreflexive.

Let R be a binary relation on a set A , R is anti-symmetric iff _____

- $a, b \in A$ if $(a, b) \in R$ and $(b, a) \in R$ then $a = b$

“-“ is a binary operation on the set of integers Z .

- True

The inverse relation R^{-1} from B to A is defined as _____

- $R^{-1} = \{(b, a) \in B \times A \mid (a, b) \in R\}$

Which of the following is always true for the matrix representation of a symmetric relation?

- Matrix is equal to its transpose

Let $A = \{1, 2, 3, 4\}$ and let R and S be transitive binary relations on A defined as; $R = \{(1, 2), (1, 3), (2, 2), (3, 3), (4, 2), (4, 3)\}$ and $S = \{(2, 1), (2, 4), (3, 3)\}$ then $R \cup S = \{(1, 2), (1, 3), (2, 1), (2, 2), (2, 4), (3, 3), (4, 2), (4, 3)\}$

- $R \cup S$ is transitive

Let $S = R$ and define the ‘square’ relation $R = \{(x, y) \mid x^2 = y^2\}$. The square is an _____ relation.

- Equivalence relation

If $x \equiv -10 \pmod{15}$. Which of the following integers are valid solution for x ?

- 5

Let R be a binary relation on a set A . If R is anti symmetric then _____

- Inverse of R is anti symmetric

If $A = \{1, 2, 3\}$ is a set and $R = \{(1, 2), (2, 2), (2, 1)\}$ is a relation on A , R is

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- ☐ **Symmetric**

Let $A=\{0,1\}$ and $B=\{1\}$. Let R and S be two binary relations on Cartesian product of A and B such that $R=\{(0,1)\}$ and $S=\{(1,1)\}$. Then R intersection $S=$ _____

- ☐ **Empty**

A relation R is said to be _____ iff it is reflexive, antisymmetric and transitive.

- ☐ **Partial order Relation**

Let $X=\{1,2,3\}$ and $Y=\{7,8,9\}$ and let f be function defined from X to Y such that f is onto then which of the following statement about f is true?

- ☐ **Co-domain of f must contain 1 element**

The function $f \circ g$ and $g \circ f$ are always equal

- ☐ **False**

If a relation $R=\{(1,2),(2,3),(3,4),(4,1),(2,2)\}$ is given then which of the following is true about this relation.

- ☐ **R is reflexive**

A set is called countable if, and only if, it is _____

- ☐ **finite**

Let $f(x) = x^2 - 1$ define function f from $\mathbb{R}$ to $\mathbb{R}$ and $c=2$ be any scalar, then $c \cdot f(x)$ is _____

- ☐ **$2x^2 + 2$**

The set $\mathbb{Z}$ of all integers is _____

- ☐ **Countable**

Let R be a binary relation on a set A . If R is anti symmetric then _____

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- Inverse of R is symmetric

For $(2x-3, 4y+2) = (1,10)$. What will be the value of x and y ?

- (2,2)

Let f and g be the two functions from R to R defined by $f(x) = |x|$ and $g(x) = \text{square root of } x^2$ for all $x \in R$. Then _____

- F(x) is not equal to g(x)

If a set A has 15 elements then $P(A)$ (power set of A) has _____ elements.

- 2^{15}

For the relation below to be a function, x cannot be what values $\{(12,14), (13,5), (-2,7), (x,13)\}$?

- X cannot be 12, 13, or -2

Let the set $A = \{1,2,3,4\}$. Then the relation $\{(2,4), (4,2)\}$ is _____

- Symmetric

For the following relation to be a function, x can not be what values?
 $R = \{(2,4), (x,1), (4,2), (5,6)\}$.

- x cannot be 2,4 and 5

Vertical line test is used to determine that whether the graph of a relation is a function or not.

- True

The properties of being symmetric and being anti symmetric are _____

- Not negative of each other

The number of elements in $A \times B$ are _____ if A is a set with '5' elements and B is a set with '4' elements.

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○ 20

$R = \{(a,1)(b,2)(c,3)(d,4)\}$ then the inverse of this relation is _____

○ $\{(1,a)(2,b)(3,c)(4,d)\}$

Logic gate NOT does not define a binary operation on $(0,1)$ because _____

○ It takes a single input and gives a single output

How many real numbers exist between 1 and 5

○ 3

The number π is

○ Irrational

The number square root 2 is

○ Irrational

Range the relation $\{(0,1)(3,22)(90,34)\}$

○ $\{0,3,90\}$

Spherical coordinate 0 is related to the cylindrical coordinate as _____

○ $@ = @$

Operation of subtraction is a binary operation on the set _____

○ Integers

Let $A = \{1,2,3,4\}$ and $R = \{(1,2),(2,3),(3,3),(3,4)\}$ be a relation on A. Then which one of the following ordered pair has made R not an irreflexive relation?

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- (3,3)

Input values of the function are called the _____

- Domain

Range of function $f(x)=|x|$ will be

- Set of positive real numbers

Which of the following is not a binary operation on the set of integers?

- Division

In the matrix representation of an irreflexive relation all the entries in the main diagonal are _____

- 0

If the partition set of A is $\{A_1, A_2\}$ then

- $A_1 \cap A_2 = \text{not empty set}$

Let $A = \{a, b\}$ then $P(A) =$

- {Non empty set, $\{a\}, \{b\}, \{a, b\}$ }

Which relations below are not functions?

- $\{(13,14), (13,5), (16,7), (18,13)\}$

In the directed graph of an antisymmetric relation there is _____ pair of arrows between two distinct elements of the set.

- No

If a relation $R = (1,1), (2,1), (2,2)$ is given then which of the following is not true about this relation

- R is irreflexive

Let R and S be transitive relations on a set A then _____

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- Neither $R \cup S$ is transitive nor $R \cap S$ is transitive

Let $R = \{(1,2)(3,4)(5,6)(7,8)\}$. Domain of the inverse of the relation is

- $\{2,4,6,8\}$

Let $A = \{1,2,3,4,5\}$ and $B = \{4,9,16,17,25\}$. Then the relation $R = \{(2,4),(3,9),(4,16),(9,17)\}$ The inverse of R is

- $\{(4,2),(9,3),(16,4),(17,9)\}$

Let R be a relation on a set A . If R is symmetric then its complement is

- Irreflexive

Which is not a binary operation on the set of natural numbers N ?

- Subtraction

If a relation $R = \{(1,1)(2,1)(1,2)(2,2)\}$ is given then which of the following is not true about this relation.

- R is irreflexive

$R = \{(a,1)(b,2)(c,3)(d,4)\}$ then the inverse of this relation is

- $\{(1,a)(2,b)(3,c)(4,d)\}$

For any set A , the Cartesian product of A and A is known as

- Universal relation

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Let $A = \{p, q, r, s\}$ and define a relation R on A by

$R = \{(p, p), (p, r), (q, r), (q, s), (r, s)\}$ Then which one of the following is the correct statement about R :

- ☐ **R is not reflexive**

$A = \{1, 2\}$ $B = \{3, 4\}$, $R = \{(1, 3), (2, 4)\}$. Then the complement of R is

_____.

- ☐ **$\{(1, 4), (2, 3)\}$**

Domain of a relation symbolically written as _____.

- ☐ **$\text{Dom}(R) = \{a \in A \mid (a, b) \in R\}$**

Let $X = \{2, 4, 5\}$ and $Y = \{1, 2, 4\}$ and R be a relation from X to Y defined by $R = \{(2, 4), (4, 1), (a, 2)\}$. For what value of 'a' the relation R is a function?

- ☐ **5**

Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, then which of the following sets represent the partition of the set A ?

- ☐ **$A = \{1, 3, 5, 7, 9\}$, $B = \{2, 4, 6\}$, $C = \{8\}$**

Let $A = \{1, 2, 3\}$ and $B = \{2, 4\}$ then number of binary relations from A to B are _____.

- ☐ **64**

A relation R is said to be _____ iff it is reflexive, antisymmetric and transitive.

- ☐ **Partial order Relation**

Let f be a function from $X = \{2, 4, 5\}$ to $Y = \{1, 2, 4, 6\}$ defined as: $f = \{(2, 6), (4, 2), (5, 1)\}$. The range of f is _____

- ☐ **$\{1, 2, 6\}$**

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Let $A=\{0,1,2\}$ and $R=\{(0,2),(1,1),(2,0)\}$ be a relation on A. Then which of the following statement about R is true?

- ☐ R is symmetric

Let $A=\{2,3,4\}$ and $B=\{2,6,8\}$ and let R be the “divides” relation from A to B i.e for all (a,b) belong to (Cartesian product of A and B), aRb iff $a \mid b$ (a divides b). Then

- ☐ $R=\{(2,2),(2,6),(2,8),(3,6),(4,8)\}$

Let $A=\{1,2,3,\dots,50\}$ and $B=\{2,4,6,8,10\}$. Then the Cartesian product of A and B has _____ elements.

- ☐ 250

In the matrix representation of an reflexive relation all the entries in the main diagonal are _____

- ☐ 1

Which of the following is not a type of a relation?

- ☐ Permutation

Let $X=\{2,4,5\}$ and $Y=\{1,2,4\}$ and R be a relation from X to Y defined by $R=\{(2,4),(4,1),(a,2)\}$. For what value of ‘a’ the relation R is a function ?

- ☐ 5

Which of the following is not a representation of a relation?

- ☐ Venn diagram

Let $A=\{1,2,3,4\}$ and define the following relations on A. Then

$R=\{(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)\}$ is _____

- ☐ R is irreflexive

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The range of $f: X \rightarrow Y$ is also called the image of

- True

Complementary Relation symbolically written as _____

- $R = A \times B - R = \{(a,b) \in A \times B \mid (a,b) \text{ not belong to } R\}$

Let $A = \{1,2,3,4\}$ and $R = (1,1)(2,2),(3,3),(4,4)$ then R is

- All options

If a relation $R = \{(1,1),(2,1),(1,2),(2,2)\}$ is given then which of the following is not true about this relation.

- R is irreflexive

Which of the following logical connective is not a binary operation ?

- Implication

A set may be divided up into its disjoint subsets, such division is called _____

- Partition

If $A = (1,2,3)$ & $B = (4,5,6)$ and $R = \{(1,4)(2,5)(3,6)(3,4)\}$ The complementary relation is _____

- $A \times B - R$ (difference or -) R

In matrix representation of a _____ relation, the diagonal entries are always 1.

- Reflexive

R is not symmetric iff there are elements a and b in A such that

- (a,b) belongs to R but (b,a) does not belong to R

Which relations below are functions?

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$$R1=\{(3,4),(4,5),(6,7),(8,9)\}$$

$$R2=\{(3,4),(4,5),(6,7),(3,9)\}$$

$$R3=\{(-3,4),(4,-5),(0,0),(8,9)\}$$

$$R4=\{(8,11),(34,5),(6,17),(8,19)\}$$

- **R1 and R3 are functions**

The logic gate OR and AND are uniary operation on $\{0,1\}$

- **False**

There is atleast one loop in the graph of an irreflexive relation

- **False**

There is atleast one loop in the graph of an reflexive relation

- **True**

A contains 3 elements and B contains 2 elements, then number of subsets of $A*B$ are _____

- **64**

Let $A=\{1,2,3\}$ and $B=\{0,1,2\}$ and $C=\{a,b\}$ $R=\{(1,0),(1,2),(3,1),(3,2)\}$
 $S=\{(0,b),(1,a),(2,b)\}$ composite of R and S= _____

- **$\{(1,b),(1,a),(3,a),(3,b)\}$**

If R is transitive then the inverse relation will be transitive

- **True**

The number of elements in the power set of P(not empty set) denoted by $P(P(\text{not empty set}))$ is

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○ 2

The function defined from $\mathbb{Z}$ to $\mathbb{Z}$ as $f(x) = 1/(x+2)(x-2)$ is not well defined because _____

○ Function is not defined at $x=2$ and $x=-2$

Rang of relation $\{(0,1),(3,22),(90,34)\}$ is _____

○ {1, 22, 34}

The number of elements in the power set of P (not empty set) denoted by P (not empty set) is

○ 1

Let $R = \{(1,2),(3,4),(5,6),(7,8)\}$. Domain of inverse of the relation is

○ {2,4,6,8}

The relation 'divides' on the set of integers is _____

○ A symmetric relation

Operation of subtraction is a binary operation on the set of _____

○ Integers

If R is transitive then the inverse relation will be transitive.

○ True

Let $A = \{1,2,3,4\}$ and define the relation R on A by

$R = \{(1,2),(2,3),(3,3),(3,4)\}$. Then _____

○ R is both reflexive and irreflexive

A set may be divided up into its disjoint subsets, such division is called

○ Partition

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If a set A contains n elements then the number of elements in its power set $P(A)$ is _____

- ☐ 2^n

Range of a relation symbolically written as _____.

- ☐ $\text{Ran } R = \{b \in B \mid (a, b) \in R\}$

Let R be a binary relation on a set A . If R is anti symmetric then _____.

- ☐ Inverse of R is anti symmetric

Let A be a set with m elements and B be a set with n elements then the number of elements in $A \times B$ are _____

- ☐ $m \cdot n$

Let $A = \{1, 2, 3, 4\}$ and define the following relation on A . Then $R = \{(1, 3), (2, 2), (2, 4), (3, 1), (4, 2)\}$ Is _____

- ☐ R is symmetric

If $A = \{1, 2, 3\}$ & $B = \{4, 5, 6\}$ and $R = \{1, 4\}, \{2, 5\}, \{3, 6\}, \{3, 4\}$. The complementary relation is _____

- ☐ $A \times B - R$

Let R be a relation on a set A . If R is reflexive then its complement is _____

- ☐ Irreflexive

$25 \equiv 1 \pmod{3}$ means that 3 divides _____

- ☐ 25-1

Let R be the universal relation on a set A then which one of the following statement about R is true ?

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- **R is reflexive, symmetric and transitive**

Domain of a relation symbolically written as _____

- **$\text{Dom}(R) = \{a \in R \mid (a,b) \in R\}$**

Let R be a relation on a set A. R is transitive if and only if for all $a,b,c \in A$ then

- **$(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$**

Let A be a non-empty set and $P(A)$ the power set of A. Define the 'subset' relation, $\subseteq$ as follows for all $X,Y \in P(A)$, $X \subseteq Y \iff$ for all x , iff $x \in X$ then $x \in Y$. Then $\subseteq$ is _____

- **$\subseteq$ is partial order relation**

Define a relation $R = \{(1,1), (2,2), (3,3), (1,3)\}$ the relation is

- **R is reflexive and transitive**

Let R be a relation on a set A. R is transitive if and only if for all $a,b,c \in A$ then.

- **$(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$**

Let R and S be reflexive relations on a set A then $R \cap S$ is reflexive

- **True**

A function whose range consists of only one element is called _____

- **One to one function**

The set Z of all integers is _____

- **Countable**

One-to-one correspondence means the condition of _____

- **Both (a) and (c)**

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Let $X=\{1,5,9\}$ and $Y=\{3,4,7\}$, Define a function f from X to Y such that $f(1)=7, f(5)=3, f(9)=$ _____. Which is true $f(9)$ to make it a one-to-one (injective) function?

- ☐ 4

What will be the fourth term of the following sequence $1/2, 2/3, 3/4$ _____?

- ☐ 4/5

The value of $6!=$

- ☐ 720

A constant function is one to one iff its _____ is a singleton.

- ☐ Domain

A constant function is onto iff its _____ is a singleton.

- ☐ Co-domain

Number of one to one functions from $X=\{a,b\}$ to $Y=\{u,v\}$ are equal to _____.

- ☐ 2

If f is defined recursively by $f(0) = -1$ and $f(n+1)=f(n)+3$, then $f(2)=$ _____.

- ☐ 5

A function whose inverse function exists is called a/an _____

- ☐ Invertible function

Let $f(2)=3$, $g(2)=3$, $f(4)=1$ and $g(4)=2$ then the value of $f \circ g(4)$ is.....

- ☐ 3

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Let $A=\{1,2,3,4\}$ and $B=\{7\}$ then the constant function from A to B is

_____.

- ☐ **Onto**

Composition of a function is a commutative operation.

- ☐ **False**

Composition of a function is not a commutative operation

- ☐ **True**

The sum of first five whole number is _____.

- ☐ **10**

If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

- ☐ **True**

Inverse of a surjective function is always a function.

- ☐ **False**

Inverse of a surjective function may not be a function

- ☐ **True**

Let $X=\{1,2,3,4\}$ and $Y=\{7,8,9\}$ and let f be function defined from X to Y such that f is onto then which of the following statement about f is true?

- ☐ **Co-domain of f must contain 3 elements**

If f; $X \rightarrow Y$ and g; $Y \rightarrow Z$ are both onto functions. Then gof; $X \rightarrow Z$ is

- ☐ **onto**

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If f and g are two one-to-one functions, then their composition $g \circ f$ is _____

- ☐ One-to-one

If $f: W \rightarrow X$, $g: X \rightarrow Y$, and $h: Y \rightarrow Z$ are functions, then _____

- ☐ $(h \circ g) \circ f = h \circ (g \circ f)$

Cardinality of positive prime numbers less than 20 is _____.

- ☐ 8

If $f(x) = \sin^{-1}(x)$ and $g(x) = \sin x$ then $g \circ f(x)$ is _____.

- ☐ X

$0! =$ _____.

- ☐ 1

An important data type in computer programming consists of _____.

- ☐ Finite sequences

Let $f(x) = 2x$ and $g(x) = x + 2$ define functions f and g from $\mathbb{R}$ to $\mathbb{R}$, then $(f \circ g)(x)$ is _____.

- ☐ $x - 2$

The total number of terms in an arithmetic series $0 + 5 + 10 + 15 + \dots + 50$ are _____.

- ☐ 11

$9!/6! =$ _____

- ☐ 504

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Let $f(x)=3x$ and $g(x)=3x-2$ define functions f and g from R to R , THEN $(F+G)(X)$ is _____

- ☐ **6x-2**

If f is a bijective function then $(f^{-1}f(x))$ is equal to

- ☐ **X**

A sequence whose terms alternate in sign is called an _____

- ☐ **Alternating sequence**

Common ration in the sequence “4, 16, 64, 256,...” is.....12.

- ☐ **4**

0.8181818181 is a infinite geometric series.

- ☐ **True**

The word ‘algorithm’ refers to a step-by-step method for performing some action.

- ☐ **True**

A predicate become _____ when its variables are given specific values

- ☐ **Sentence**

The sum of two irrational numbers must be an irrational number.

- ☐ **False**

The sum of two irrational numbers need not be irrational number

- ☐ **True**

The division by zero is allowed in mathematics.

- ☐ **Fasle**

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The product of any two consecutive positive integers is divisible by 2

- ☐ True

If 'n' is an odd integer then $n^3 + n$ is _____.

- ☐ Even

For integers a,b,c, If divides b and a divides c, then a divides (a+b).

- ☐ False

Quotient remainder theorem states that for any positive integer d, there exist unique integer q and r such that _____ and $0 \leq r < d$

- ☐ $N = d \cdot q + r$

A rule that assigns a numerical value to each outcome in a sample space is called

- ☐ Random variable

If A and B are two disjoint (mutually exclusive) events then $P(A \cap B) =$

- ☐ $P(A) + P(B)$

How many ways are there to select five players from a 10 member tennis team to make a trip to a match to another school?

- ☐ $C(10,5)$

The expectation of x is equal to

- ☐ $\sum x f(x)$

If $P(A \cap B) = P(A) P(B)$ THEN THE events A and B are called

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- **Independent**

A walk that starts and ends at the same vertex is called.

- **None optins**

How many integers from 1 through 1000 are neither multiple of 3 nor multiple of 5

- **497**

What is the probability of getting a number greater than 4 when a die is thrown?

- **3/5**

Eater formula for graphs is _____.

- **$F=e-v+2$**

$X+a, x+3a, x+5a, \dots$ is an _____

- **Arithmetic sequence**

Composition of a function is a commutative operation.

- **False**

Real valued function is a function that assigns _____ to each member of its domain.

- **Only a real number**

Let $X=\{1,2,3,4\}$ and a function 'f' defined on X $f(1)=1, f(2)=2, f(3)=3, f(4)=4$ then _____

- **F is an identity function**

A constant function is surjective if and only if _____

- **The co-domain consists of a single element**

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Cardinality means the total number of elements in a set.

- ☐ True

If $f(x)=2x$ and $g(x)=x$ then $g(f(x))$ is _____.

- ☐ $2x^2$

Let $f:R \rightarrow R$ is one to one function then c, f, c is not equal 0 is also one to one function.

- ☐ True

Let $X=\{1,2,3,4\}$ and a function 'f' defined from X to X by $f(1)=1, f(2)=1, f(3)=1, f(4)=1$ then which of the following is true?

- ☐ F is a constant function

If f and g are two one-to-one functions, then their composition that is $g \circ f$ is one-to-one.

- ☐ True

Which of the following is not correct for a 'sequence'?

- ☐ A sequence is a relation whose domain is the set of natural numbers

$F(x)=x^2$ is not one to one function from R to R^+

- ☐ True

Let $f: R \rightarrow R$ is one to one function then c, f, c is not equal to is also one to one function.

- ☐ True

Let $f(x)=x+2$ then $f^{-1}(x)$ is _____

- ☐ $x-2$

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Let $f(x)=x^2+1$ define functions f from $\mathbb{R}$ to $\mathbb{R}$ and $c=2$ be any scalar, then $c, f(x)$ is _____.

- ☐ **$2x^2-1$**

One to one correspondence means the condition of _____.

- ☐ **Both (a) and (c)**

A function $F: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \text{square root } x$ is a real valued function.

- ☐ **False**

If $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x)=e^x$ is a real valued function of a real variable.

- ☐ **True**

A function $F: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \log x$ is a real valued function

- ☐ **True**

$\frac{1}{2}$, then 3rd term of sequence is _____.

- ☐ **$\frac{1}{2}$**

The process of defining an object in terms of smaller versions of itself is called recursion.

- ☐ **True**

Which of the following is not correct for a 'sequence'?

- ☐ **A sequence is a relation whose domain is the set of natural numbers**

A set is called countable if, and only if, it is _____.

- ☐ **Finite and countable infinite....both**

A set that is not countable is called _____.

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- ☐ **Uncountable**

A sequence whose terms alternate in sign is called an _____.

- ☐ **Alternating sequence**

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is one to one function then c, f is also one to one function for _____.

- ☐ **C is not equal 0**

Let $f(x) = x+3$ then $f^{-1}(x)$ is _____.

- ☐ **x-3**

Let f and g be two functions defined by $f(x) = x+2$ and $g(x) = 2x+1$. Then the composition of f and g is _____.

- ☐ **2x+3**

Number of one to one functions form $X = \{a, b\}$ to $Y = \{u, v\}$ are equal to _____.

- ☐ **2**

The fibonacci sequence is deined as $F_0=1, F_1=1, F_k=F_{k-1}+F_{k-2}$ for all integers $k \geq 2$ then which of the following is true for F_2

- ☐ **$F_2 - F_1 = 2 + 1 = 3$**

$x+a, x+3a, x+5a, \dots$ is a/an _____.

- ☐ **Arithmetic sequence**

Inverse of a function may not be a function.

- ☐ **True**

In the following sequence $a_k = K/(k+1)$, for $k=1$, a_1 will be _____.

- ☐ **$\frac{1}{2}$**

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If $f(x)=x$ and $g(x)= -x$ are both one to one function then $(f+g)(x)$ is also one to one function.

- ☐ False

If $f(x)=x$ and $g(x)= -x$ are both one to one function then $(f+g)(x)$ is not one to one function.

- ☐ True

The function 'f' and 'g' are inverse of each other if and only if their composition gives _____.

- ☐ Identity function

Which of the following set is the domain of a sequence?

- ☐ Set of real numbers

Let C is defined as the set of all countries in the world then C is a _____.

- ☐ Finite set

A constant function is surjective if and only if _____.

- ☐ The co domain consists of a single element

The sum of the series $a_1 + a_2 + a_3 + \dots$ can be written as _____.

- ☐ $\sum_{i=1}^{\infty} a_k$

Inverse of a function may not be a function.

- ☐ True

If $_F_k = F_k - 1 + F_k - 2$ then $_F_0 = 1, _f_1 = 2$, then $_F_2 =$ _____

- ☐ 3

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Let $X=\{1,2,3,4\}$ and a function 'f' defined from X to X by $f(1)=1$, $f(2)=1$, $f(3)=1$, $f(4)=1$ then which of the following is true?

- ☐ F is a constant function

The composition of function is always

- ☐ Associative

A set is countably infinite if, and only if, it has the same cardinality as the set of

- ☐ Positive integers

Two functions 'f' and 'g' from 'X' to 'Y' are said to be equal if and only if _____.

- ☐ $F(x)=g(x)$ for all 'x' belongs to X

$Y=x^3$ is a graph of bijective function from R to R.

- ☐ True

Domain and range are same for_____.

- ☐ Identity function

Let $X=\{1,2,3,4\}$ and a function 'f' defined on X by $f(1)=1, f(2)=2, f(3)=3$

- ☐ F is an identity function

Composition of a function is a commutative operation.

- ☐ True

Inverse of a surjective function is always a function.

- ☐ False

A function whose inverse function exists is called a/an_____.

- ☐ Invertible

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Given a set X define a function I from X to X by $i(x)=x$ from all x belonging to X . Then _____.

- ☐ **I is both injective and surjective**

Let $f: R \rightarrow R$ is a one to one function then c, f is also one to one function for.

- ☐ **C is not equal 0**

Let $X=\{1,5,9\}$ and $Y=\{3,4,7\}$. Define a function f from X to Y such that $f(1)=7, f(5)=3, f(9)=$ _____. Which is true for $f(9)$ to make it a one-to-one (injective) function?

- ☐ **4**

Which of the following is not a predecessors of a_k ?

- ☐ **A_{k+1}**

Two functions ' f ' and ' g ' from X to Y are said to be equal if and only if _____.

- ☐ **$F(x)=g(x)$ for all ' x ' belongs to x**

The two functions ' f ' and ' g ' are equal if _____.

- ☐ **$F(x) = 3x$ and $g(x) = 6x^2 + 3x/2x^2 + 1$ for all $x \in R$**

If first term of a geometric sequence is 2 and common ratio is $\frac{1}{2}$, then 3rd ter, of sequence.

- ☐ **$\frac{1}{4}$**

$Y = \text{square root } x$ is an _____ function form R^+ to R

- ☐ **One to one function**

$Y = x^2$ is an _____ function form R to R^+

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- ☐ NOT ONE TO ONE FUNTION

If a function $(g \circ f)(x) : X \rightarrow Z$ is defined as $(g \circ f)(x) = g(f(x))$ for all $x \in X$, Then the function _____.

- ☐ (gof)

If 0 is the first term and -2 be the common difference of an arithmetic series, then the sum of first five terms of series is _____.

- ☐ -20

If $f(x) = \sin^{-1}(x)$ and $g(x) = \sin x$ then $g \circ f(x)$ is _____.

- ☐ X

$f: X \rightarrow Y$ that is both one to one and onto is called a _____.

- ☐ Bijective function

What does 'y' denotes in a geometric sequence?

- ☐ Common ratio

Let g be a function defined by $g(x) = x + 1$. Then the composition of $(g \circ g)$.

- ☐ $X + 2$

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

- ☐ True

Which of the following is true for the following sequence?

- ☐ If n is even, then $C_n = 2$ and if n is odd, then $C_n = 0$

The function 'f' and 'g' are inverse of each other if and only if their composition gives _____.

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- Identity function

$N!$ is defined to be _____.

- The product of the integers from 1 to n

Let $A = \{1,2,3,4\}$ and $B = \{7\}$ then the constant function from A to B

- Both one to one and onto

A set is called countable if, and only if, it is _____.

- Countably infinite and finite

If $f(x) = x$ and $g(x) = -x$ are both one to one functions then $(f+g)(x)$ is also one to one function.

- False

If a is the 1st term and d be the common difference of an arithmetic sequence then the sequence is $a, a+d, a+2d, a+3d, \dots$

- True

$x+a, x+3a, x+5, \dots$ is a/n _____.

- Arithmetic sequence

inverse of an injective function may not be a function.

- True

$y=x^3$ is a graph of bijective function from R to R

- False

Two functions ' f ' and ' g ' from x to y are said to be equal if and only if _____.

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- $F(x)=g(x)$ for all 'x' belongs to x

Common ration in sequence '36, 12, 4 , 4/3,' is

- $1/3$

A set is called finite if, and only if, is the _____ or there is -----.

- Empty set or one-to-one

Let $f(X)=x^2-1$ and $g(x)=x+1$ define functions f and g from R to R, then $(f/g)_x$

- $x-1$

A graph of a function f is one to one iff every horizontal line intersects the graph in at most one point.

- True

Let g be a function defined by $g(x) = x+1$. Then the composition of (gog).

- $X+2$

$F:X \rightarrow Y$ that is both one to one and onto is called a _____.

- Bijective function

What does 'r' denotes in a geometric sequence?

- Common ratio

Let $A=\{1,2,3,4\}$ and $B=\{7\}$ then the constant function from A to B is _____.

- Both one to one and onto

$N!$ is defined to be _____.

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- **The product of the integers from 1 to n**

A set is called countable if, and only if, it is _____.

- **Both b and c**

Which of the following is the example of an alternating sequence?

- **$C_n = n/n+1$ for $n \geq 0$**

$X+a, x+3a, x+5a, \dots$ is an _____

- **Arithmetic sequence**

0, -5, -10, -15, ... is an _____.

- **Arithmetic sequence**

5, 9, 13, 17, ... is an _____.

- **Arithmetic sequence**

An important data type in computer programming consists of _____.

- **Finite sequence**

One-to-one correspondence means the condition of _____.

- **One-one and ontoboth**

Cardinality means the total number of elements in a set.

- **True**

Inverse of a function may not be a function.

- **True**

The functions 'f' and 'g' are inverse of each other if and only if their composition gives _____.

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- Identity function

A set is called finite if , and only if, it is the _____ or there is _____.

- Empty set or one-to-one

Let f and g be the two functions from R to R defined by $f(x) = |x|$ and $g(x) = \text{square root } x^2$ for all $x \in R$, then _____.

- $F(x)$ is not equal to $g(x)$

If $f^{-1}(x) = 6 - x/2$ then $f^{-1}(2)$ is _____.

- 2

Let $f(2)=3$, $g(2)=3$, $f(4)=1$ and $g(4)=2$ then the value of $f \circ g(4)$ is _____.

- 3

An important data type in computer programming consists of _-----.

- Finite sequence

Let $f(x)=x+3$ then $f^{-1}(x)$ is _____.

- $x-3$

If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are both onto function. Then $g \circ f: X \rightarrow Z$ is _____.

- One-to-one function

The functions ' f ' and ' g ' are inverse of each other if and only if their composition gives

- Identity function

The two function ' f ' and ' g ' are equal if _____.

- $F(x) = 3x$ and $g(x) = 6x^2 + 3x/2x + 1$ for all $x \in R$

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Two functions 'f' and 'g' from X To Y are said to be equal if and only if--
----.

- ☐ $F(x)$ and $g(x)$ for all 'x' belongs to X

Which of the following set is the domain of a sequence?

- ☐ Set of natural numbers

If 1st term of a geometric sequence is 2 and common ratio is $1/2$, then 3rd term of sequence is _____.

- ☐ $1/2$

Composition of a function is a commutative operation.

- ☐ True

If a function $(g \circ f)(x): X \rightarrow Z$ is defined as $(g \circ f)(x) = g(f(x))$ for all $x \in X$. Then the function _____ is known as composition of f and g.

- ☐ $(g \circ f)$

A set is countably infinite if and only if, it has the same cardinality as the set of _____.

- ☐ Positive integers

The 3rd term of the sequence $b_n = 5^n$ is _____.

- ☐ 125

If f is a bijective function then $(f^{-1}(f(x)))$ is equal to _____.

- ☐ X

Let $f(x) = x$ and $g(x) = -x$ for all $x \in \mathbb{R}$, then $(f+g)(x)$ is _____.

- ☐ 0

An infinite sequence may have only a finite number of values.

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☐ **True**

The functions $f \circ g$ and $g \circ f$ are always equal .

☐ **False**

If f and g are two one-to-one functions, then their composition that is $g \circ f$ is one-to-one.

☐ **True**

A function whose range consists of only one element is called _____.

☐ **Constant function**

Let $X = \{1, 5, 9\}$ and $Y = \{3, 4, 7\}$. Define a function f from X to Y such that $f(1)=7$, $f(5)=3$, $f(9)=4$ then which of the following statement about ' f ' is true?

☐ **f is both one-to-one and onto**

$Y = x^3$ is a graph of bijective function from $\mathbb{R}$ to $\mathbb{R}$.

☐ **True**

A function whose inverse function exists is called a/an _____.

☐ **Invertible**

Let f and g be two functions defined by $f(x) = x+2$ and $g(x) = 2x+1$. Then the composition of f and g is _____.

☐ **$2x + 5$**

If r is a positive real number, then the value of r in $3, r, r = -27r$ is

☐ **-9**

The _____ of the terms of a sequence forms a series.

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☐ Sum

The sum of first five whole number is _____.

☐ 10

If $f_k = f_{k-1} + f_{k-2}$ then $f_0 = 1$, $f_1 = 2$, then $f_2 =$ _____.

☐ 3

Let $f(x) = 3x$ and $g(x) = x + 2$ define functions f and g from R to R , then $(f.g)(x)$ is _____.

☐ $3x^2 + 6x$

Let R be a relation on a set A . If R is reflexive then its compliment is _____.

☐ Irreflexive

If $A =$ Set of students of virtual university then A has been written in the _____.

☐ Descriptive form

If a function $(g \circ f)(x): X \rightarrow Z$ is defined as $(g \circ f)(x) = g(f(x))$ for all $x \in X$. Then the function _____ is known as composition of f and g .

☐ $(g \circ f)$

If X and Y are independent random variables and a and b are constants, then $\text{Var}(aX + bY)$ is equal to

☐ $a\text{Var}(X) + b\text{Var}(Y)$

Let $A = \{1, 2, 3\}$ and $B = \{2, 4\}$ then number of functions from A to B are _____.

☐ 8

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p is equivalent to q' means _____.

- **p is necessary and sufficient for q.**

Let A and B be subsets of U with $n(A) = 12$, $n(B) = 15$, $n(A') = 17$, and $n(A \cap B) = 8$, then $n(U) =$ _____.

- **29**

For the following relation to be a function, x can not be what values?
 $R = \{(2,4), (x,1), (4,2), (5,6)\}$

- **x cannot be 2, 4 or 5**

Find the number of the word that can be formed of the letters of the word "ELEVEN".

- **120**

There are three bus lines between A and B, and two bus lines between B and C. Find the number of ways a person can travel round trip by bus from A to C by way of B?

- **6**

Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog, how many jog?

- **16**

A predicate becomes _____ when its variables are given specific values.

- **statement**

Find the number of distinct permutations that can be formed using the letters of the word "BENZENE"

- **420**

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Suppose there are 8 different tea flavors and 5 different biscuit brands. A guest wants to take one tea and one brand of biscuit. How many choices are there for this guest?

- ☐ 40

In how many ways a student can choose one of each of the courses when he is offered 3 mathematics courses, 4 literature courses and 2 history courses.

- ☐ 24

If $p \leftrightarrow q$ is True, then _____.

- ☐ p and q both are True.

If A and B be events with $P(A) = 1/3$, $P(B) = 1/4$ and $P(A \cap B) = 1/6$, then $P(A \cup B) =$ _____.

- ☐ 5/12

an integer n is a perfect square if and only if _____ for some integer k.

- ☐ $n = k^2$

If A and B are disjoint finite sets then $n(A \cup B) =$ _____.

- ☐ $n(A) + n(B)$

Let $X = \{2, 4, 5\}$ and $Y = \{1, 2, 4\}$ and R be a relation from X to Y defined by $R = \{(2,4), (4,1), (a,2)\}$. For what value of 'a' the relation R is a function ?

- ☐ 5

$\sim(P \rightarrow q)$ is logically equivalent to _____.

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- $p \wedge \sim q$

A tree is normally constructed from _____.

- left to right

A Random variable is also called a _____.

- Chance Variable

The conjunction $p \wedge q$ is True when _____.

- p is True, q is True

The logical statement $p \wedge q$ means _____.

- p AND q

Which of the followings is the factorial form of 5 . 4?

- 5!/3!

What is the minimum number of students in a class to be sure that two of them are born in the same month?

- 13

If p is false and q is true, then $\sim p \leftrightarrow q$ is _____.

- True

If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

- TRUE

$(p \vee \sim p)$ is the _____.

- Tautology

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$(-2)! = \underline{\hspace{2cm}} ?$

- **Undefined**

If $p =$ It is raining, $q =$ She will go to college
"It is raining and she will not go to college"
will be denoted by

- **$p \wedge \sim q$**

Let $X = \{1, 2, 3\}$, then 2-combinations of the 3 elements of the set X are
 $\underline{\hspace{2cm}} ?$

- **$\{1, 2\}, \{1, 3\}$ and $\{2, 3\}$**

Let $f(x) = x^2 + 1$ define functions f from R to R and $c = 2$ be any scalar,
then $c.f(x)$ is $\underline{\hspace{2cm}}$.

- **$2x^2 + 2$**

The disjunction of p and q is written as $\underline{\hspace{2cm}}$.

- **$p \vee q$**

If X and Y are independent random variables, then $E(XY)$ is equal to

- **$E(x)E(y)$**

How many possible outcomes are there when a fair coin is tossed four times?

- **16**

Which of the followings is the product set $A * B * C$? where $A = \{a\}$, $B = \{b\}$, and $C = \{c, d\}$.

- **$\{(a, b, c), (a, b, d)\}$**

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The number of the words that can be formed from the letters of the word, "COMMITTEE" are

- **$9! / (2!2!2!)$**

One-to-One correspondence means the condition of _____.

- **One-One and onto**

The functions $f \circ g$ and $g \circ f$ are always equal.

- **FALSE**

If order matters and repetition is allowed, then which counting method should be used in order to select 'k' elements from a total of 'n' elements?

- **K-Sample**

Determine values of x and y , where $(2x, x + y) = (8, 6)$.

- **$x = 4$ and $y = 2$**

Let g be a function defined by $g(x) = x + 1$. Then the composition of $(g \circ g)(x)$ is _____.

- **$x + 2$**

What is the truth value of the sentence?
'It rains if and only if there are clouds.'

- **False**

'Reductio ad absurdum' is another name of _____.

- **Proof by contradiction**

X belongs to A or x belongs to B , therefore x belongs to _____.

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- **A union B**

Which of the followings is the product set $A * B * C$? Where $A = \{a\}$, $B = \{b\}$, and $C = \{c, d\}$.

- **$\{(a, b, c), (a, b, d)\}$**

Real valued function is a function that assigns _____ to each member of its domain.

- **Only a real number**

The negation of "Today is Friday" is

- **Today is not Friday**

A non-zero integer d divides an integer n if and only if there exists an integer k such that _____.

- **$n = d k$**

The statement $p \rightarrow q$ is logically equivalent to $\sim q \rightarrow \sim p$

- **True**

Let R be the universal relation on a set A then which one of the following statement about R is true?

- **R is reflexive, symmetric and transitive.**

Let $f(x)=3x$ and $g(x) = 3x - 2$ define functions f and g from R to R . Then $(f+g)(x) =$ _____.

- **$6x - 2$**

The switches in parallel act just like _____.

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- **OR gate**

The converse of the conditional statement $p \rightarrow q$ is

- **$q \rightarrow p$**

If X and Y are random variables, then $E(aX)$ is equal to

- **$aE(X)$**

Which of the following statements is true according to the Division Algorithm?

- **$17 = 5 \times 3 + 2$**

Let $p \rightarrow q$ be a conditional statement, then the statement $q \rightarrow p$ is called _____.

- **Converse**

The disjunction $p \vee q$ is False when _____.

- **p is False, q is False.**

A student can choose a computer project from one of the two lists. The two lists contain 12 and 18 possible projects, respectively. How many possible projects are there to choose from?

- **30**

The converse of the conditional statement 'If I live in Quetta, then I live in Pakistan' is _____.

- **If I live in Pakistan, then I live in Quetta.**

The functions ' f ' and ' g ' are inverse of each other if and only if their composition gives _____.

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- Identity function

$P(0, 0) = \underline{\hspace{2cm}}?$

- 1

Let p_1, p_2, p_3 be True premises in a given Truth Table. If the conjunctions of the Conclusion with each of p_1, p_2, p_3 are True, then the argument is .

- Valid

If p is false and q is false, then $\sim p$ implies q is .

- False

A box contains 5 different colored light bulbs. Which of the followings is the number of ordered samples of size 3 with replacement?

- 125

Let $A = \{2, 3, 5, 7\}$, $B = \{2, 3, 5, 7, 2\}$, $C =$ Set of first five prime numbers. Then from the following which statement is true ?

- $A = B$

The set of prime numbers is .

- Infinite set

The contrapositive of the conditional statement 'If it is Sunday, then I go for shopping' is .

- I do Not go for shopping, then it is Not Sunday.

Let p be True and q be True, then $(\sim p \wedge q)$ is .

- False

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In how many ways a student can choose a course from 2 science courses, 3 literature courses and 5 art courses.

- ☐ **30**

The method of loop invariants is used to prove _____ of a loop with respect to certain pre and post-conditions.

- ☐ **correctness**

A student is to answer five out of nine questions on exams. Find the number of ways that can choose the five questions

- ☐ **126**

If A and B are any two sets, then $A - B = B - A$

- ☐ **False**

There are 5 girls students and 20 boys students in a class. How many students are there in total?

- ☐ **25**

